```
import pandas as pd
import zipfile

zip_file_path = 'fake_job_postings.csv.zip'
csv_file_path = 'fake_job_postings.csv'

with zipfile.ZipFile(zip_file_path, 'r') as zip_ref:
    zip_ref.extractall('.')

df = pd.read_csv(csv_file_path)
df.head()
```

	job_id	title	location	department	salary_range	company_profile	descri
0	1	Marketing Intern	US, NY, New York	Marketing	NaN	We're Food52, and we've created a groundbreaki...	Food52, growing, James Award-
1	2	Customer Service - Cloud Video Production	NZ, , Auckland	Success	NaN	90 Seconds, the worlds Cloud Video Production ...	Organised - Foc Vibrant - Aweso
2	3	Commissioning Machinery Assistant (CMA)	US, IA, Wever	NaN	NaN	Valor Services provides Workforce Solutions th...	Our client, loc Houston, is a
3	4	Account Executive - Washington DC	US, DC, Washington	Sales	NaN	Our passion for improving quality of life thro...	THE COMPANY – Environ Systems F
4	5	Bill Review Manager	US, FL, Fort Worth	NaN	NaN	SpotSource Solutions LLC is a Global Human Cap...	JOB Itemization F ManagerLOCAT

```
df.info()
df.columns
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 17880 entries, 0 to 17879
Data columns (total 18 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   job_id               17880 non-null  int64
 1   title                17880 non-null  object
 2   location             17534 non-null  object
 3   department           6333 non-null   object
 4   salary_range         2868 non-null   object
 5   company_profile      14572 non-null  object
 6   description          17879 non-null  object
```

```
 7   requirements         15184 non-null  object
 8   benefits             10668 non-null  object
 9   telecommuting        17880 non-null  int64
 10  has_company_logo     17880 non-null  int64
 11  has_questions        17880 non-null  int64
 12  employment_type      14409 non-null  object
 13  required_experience  10830 non-null  object
 14  required_education   9775 non-null   object
 15  industry             12977 non-null  object
 16  function             11425 non-null  object
 17  fraudulent           17880 non-null  int64
dtypes: int64(5), object(13)
memory usage: 2.5+ MB
Index(['job_id', 'title', 'location', 'department', 'salary_range',
       'company_profile', 'description', 'requirements', 'benefits',
       'telecommuting', 'has_company_logo', 'has_questions', 'employment_type',
       'required_experience', 'required_education', 'industry', 'function',
       'fraudulent'],
      dtype='object')
```

```
df = df[['description', 'fraudulent']]
df = df.dropna()

X = df['description']
y = df['fraudulent']
```

```
from sklearn.feature_extraction.text import TfidfVectorizer

vectorizer = TfidfVectorizer(stop_words='english')
X_vectorized = vectorizer.fit_transform(X)
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X_vectorized, y, test_size=0.2
)
```

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()
model.fit(X_train, y_train)
```

```
▾ LogisticRegression
LogisticRegression()
```

```
from sklearn.metrics import accuracy_score

y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.9628076062639821
```

```
sample = ["Earn money fast. No experience required. Work from home"]
```

```
sample_vec = vectorizer.transform(sample)
print(model.predict(sample_vec))
```

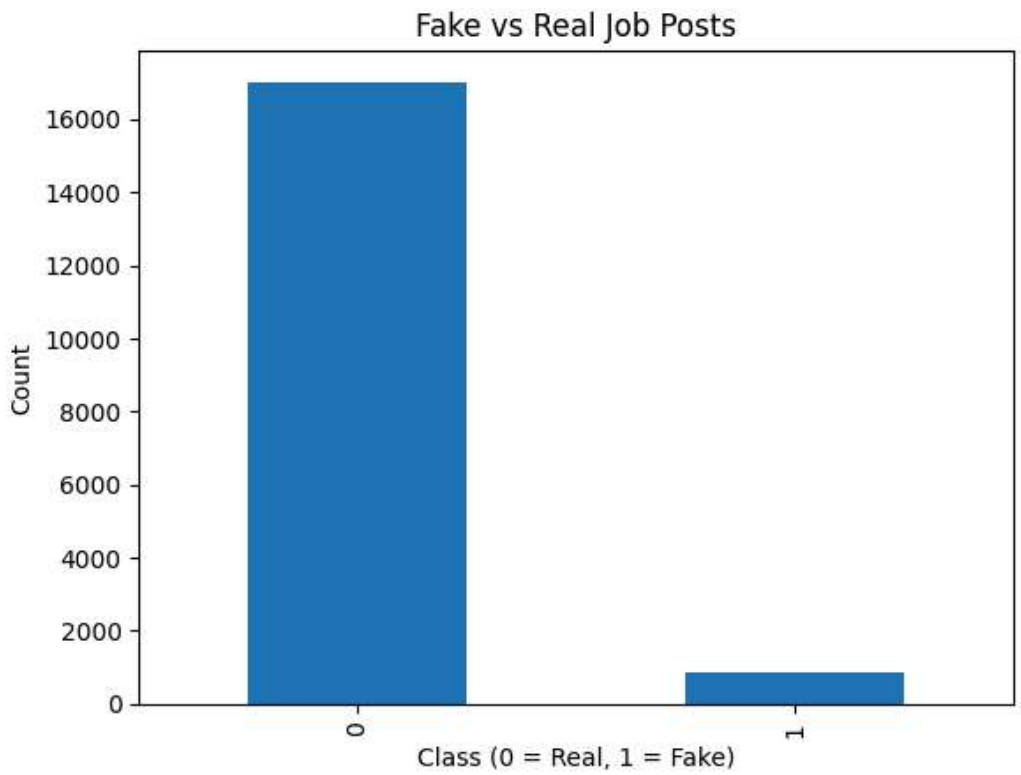
```
[1]
```

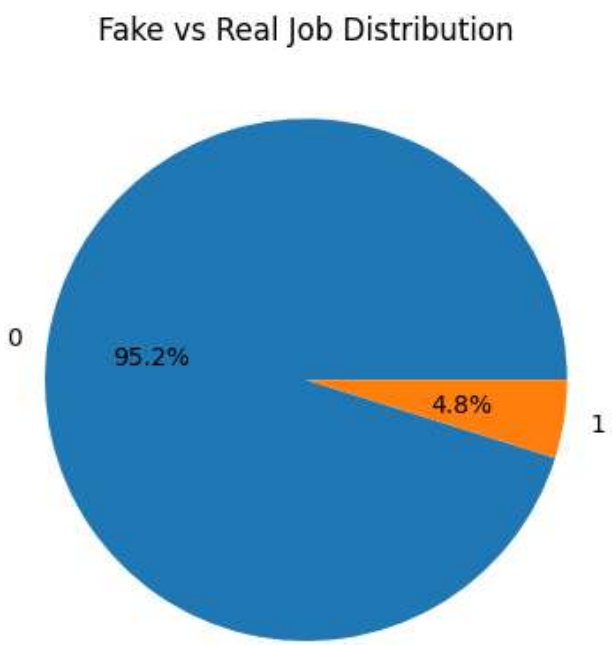
```
import pickle

with open('model.pkl', 'wb') as f:
    pickle.dump(model, f)

with open('vectorizer.pkl', 'wb') as f:
    pickle.dump(vectorizer, f)
```

```
import os

try:
    print("Model size:", os.path.getsize('model.pkl'), "bytes")
except FileNotFoundError:
    print("Error: model.pkl not found. Please ensure the model was saved correctly in the pr

try:
    print("Vectorizer size:", os.path.getsize('vectorizer.pkl'), "bytes")
except FileNotFoundError:
    print("Error: vectorizer.pkl not found. Please ensure the vectorizer was saved correctly
```

```
Error: model.pkl not found. Please ensure the model was saved correctly in the previous step.
Error: vectorizer.pkl not found. Please ensure the vectorizer was saved correctly in the prev
```

```
import matplotlib.pyplot as plt
import pandas as pd
import zipfile
import os

zip_file_path = 'fake_job_postings.csv.zip'
csv_file_path = 'fake_job_postings.csv'

# Check if the zip file exists, if not, download it
if not os.path.exists(zip_file_path):
    print(f"Downloading {zip_file_path}...")
    # Using a known public direct download link for the dataset
    !wget -q https://github.com/sujitmandal/fake_job_postings_dataset/raw/main/fake_job_post

# Ensure the zip file is extracted
with zipfile.ZipFile(zip_file_path, 'r') as zip_ref:
    zip_ref.extractall('.')

df = pd.read_csv(csv_file_path)
df = df[['description', 'fraudulent']]
df = df.dropna()

df['fraudulent'].value_counts().plot(kind='bar')
plt.title("Fake vs Real Job Posts")
plt.xlabel("Class (0 = Real, 1 = Fake)")
plt.ylabel("Count")
plt.show()
```




```
df['fraudulent'].value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.title("Fake vs Real Job Distribution")
plt.ylabel("")
plt.show()
```